

RESEARCH ARTICLE AND REPRODUCIBLE PROOF AUDIT

A Laurent-series case of Gonshor's birthday conjecture

Exact ordinal lengths, finite products, and infinite tails

Prepared for Vladimir Reshetnikov

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Abstract

Gonshor's conjecture asks whether the birthday of a product of surreal numbers is bounded by the natural product of their birthdays. We give a detailed proof of this inequality on the canonically embedded Laurent-series field $\mathbb{R}((\omega^{-1}))$. The argument first derives an exact birthday formula for finite Laurent polynomials. The only information required from their bounded part is the lowest exponent, its coefficient's birthday, and one dyadic versus nondyadic boundary condition. A positive-polynomial/bounded-tail decomposition then proves the product estimate for finite Laurent polynomials. A finite-jet stabilization argument transfers this estimate to arbitrary Laurent series, without assuming continuity of natural ordinal multiplication. We also prove the conjecture for finite normal forms with ordinal exponents, determine the birthday of every infinite Laurent series, and obtain exact birthdays of a class of reciprocals. An exact-arithmetic program independently checks the finite formula and explores fractional dyadic exponents beyond the proved case.

Status and scope. This is a partial result, not a resolution of Gonshor's conjecture for all surreal numbers. The proofs below use the standard normal-form and sign-expansion theorems, with the subsequent arguments supplied in detail. They have not been independently refereed or checked by a proof assistant. The literature search located the documented conjecture and its known monomial case, but did not establish priority for the Laurent-series restriction proved here; no novelty claim is made.

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1 The problem and the result

A surreal number can be represented by a sequence of + and – signs indexed by an ordinal. Its *birthday*, written $b(x)$, is the length of this sequence; in particular $b(0) = 0$. For ordinals α, β , let $\alpha \oplus \beta$ and $\alpha \otimes \beta$ denote their Hessenberg natural sum and natural product. The problem selected for this investigation is the following.

Conjecture 1.1 (Gonshor’s birthday product conjecture). For every $x, y \in \mathbf{No}$,

$$b(xy) \leq b(x) \otimes b(y). \quad (1.1)$$

Gonshor proposed the inequality in his 1986 book, p. 96 [1]. Van den Dries and Ehrlich proved it when both factors are single normal-form terms, $x = r\omega^a$ and $y = s\omega^b$, where r, s are real and a, b are arbitrary surreal numbers [2, Proposition 3.6]. Their general estimate is the weaker bound

$$b(xy) \leq \omega \otimes b(x) \otimes b(y) \otimes b(y) \quad (1.2)$$

[2, Corollary 4.3]. In an August 2024 discussion, Ehrlich reported that he knew of no improvement of the affirmative monomial case [6]. Searches conducted for this article on 20 September 2026 did

not locate a later resolution. That is evidence for selecting the problem, not a proof of its current worldwide research status.

The principal result is a restriction of (1.1) to a familiar, non-Archimedean field with unrestricted real coefficients.

Theorem 1.2 (Laurent-series restriction). *Let*

$$\mathcal{L} = \mathbb{R}((\omega^{-1})) = \left\{ \sum_{k \leq N} c_k \omega^k : N \in \mathbb{Z}, c_k \in \mathbb{R} \right\} \subseteq \mathbf{No}, \quad (1.3)$$

where the sum is interpreted as a surreal normal form. Then

$$b(xy) \leq b(x) \otimes b(y) \quad (x, y \in \mathcal{L}).$$

The notation includes finite Laurent polynomials and series with infinitely many nonzero terms. These are formal Hahn series, not sums taken in the ordinary real topology. Integer exponents are bounded above, so every interval of exponents bounded on both sides contains only finitely many possible terms. This discreteness is crucial to the passage from finite to infinite products.

Two further results clarify the mechanism. For finite normal forms whose exponents are ordinals, birthdays add by natural addition term by term (Theorem 4.1). For an infinite Laurent series, however,

$$b(x) = b(x_{>0}) + \omega^2, \quad (1.4)$$

where the plus on the right is *ordinary ordinal addition* (Theorem 7.1). Confusing that plus with $\oplus$ would invalidate a tempting direct extension of the finite proof.

1.1 A guide to the proof

Sections 2–3 establish the notation and the sign-expansion input. Section 4 handles ordinal-exponent polynomials. Section 5 computes finite Laurent birthdays exactly. Section 6 proves the finite product estimate by separating positive powers from the bounded tail. Section 7 passes to infinite series by preserving a fixed bound for every finite product truncation. Sections 8–10 give consequences, failed generalizations, and reproducible checks.

2 Three arithmetic conventions

The same objects occur both as surreal numbers and as ordinal lengths. We keep the associated operations separate.

2.1 Ordinary and natural ordinal arithmetic

Unadorned $+$ between birthdays denotes *ordinary ordinal addition*, the operation corresponding to concatenation of well-ordered sequences. Thus

$$1 + \omega = \omega, \quad \omega + \omega^2 = \omega^2, \quad \omega^2 + \omega > \omega^2.$$

The symbols $\oplus$ and $\otimes$ always mean natural operations. If

$$\alpha = \sum_i \omega^{\xi_i} a_i, \quad \beta = \sum_i \omega^{\xi_i} b_i \quad (\xi_0 > \xi_1 > \cdots)$$

are Cantor normal forms written over the same finite list of exponents, allowing zero coefficients, then

$$\alpha \oplus \beta = \sum_i \omega^{\xi_i} (a_i + b_i).$$

Natural multiplication is determined by distributivity and

$$\omega^\xi \otimes \omega^\eta = \omega^{\xi \oplus \eta}. \quad (2.1)$$

Both operations are associative and commutative. Natural addition is strictly increasing in either variable, and natural multiplication is strictly increasing in either variable when the other is nonzero. In particular,

$$(\alpha \oplus \beta) \otimes (\gamma \oplus \delta) = \alpha \otimes \gamma \oplus \alpha \otimes \delta \oplus \beta \otimes \gamma \oplus \beta \otimes \delta.$$

These properties also follow from the Cantor-normal-form definitions, comparing the largest exponent at which two expressions differ.

For ordinals below ω^ω , natural arithmetic is simply polynomial arithmetic in a formal ω , with nonnegative integer coefficients. For example,

$$(\omega \cdot 2 + 3) \otimes (\omega + 4) = \omega^2 \cdot 2 + \omega \cdot 11 + 12.$$

In an ordinal expression, $\omega^k \cdot n$ means ordinary right multiplication by the finite integer n ; it agrees with $\omega^k \otimes n$.

2.2 Surreal arithmetic and normal forms

In a surreal expression, such as $x + y$ or $\omega^a \omega^b$, the arithmetic is that of the surreal field. The omega map obeys

$$\omega^a \omega^b = \omega^{a \oplus b}, \quad (2.2)$$

where the exponent sum is surreal addition. When a, b are ordinals this surreal sum is their natural ordinal sum $a \oplus b$, not their ordinary sum. When they are integers it is their usual integer sum.

We use the standard normal-form theorem: a surreal is represented uniquely by

$$x = \sum_{i < \lambda} r_i \omega^{a_i}, \quad a_0 > a_1 > \cdots, \quad r_i \in \mathbb{R} \setminus \{0\},$$

with reverse-well-ordered support. Addition and multiplication agree with Hahn series operations [1, 5]. All subsequent infinite series have integer exponents bounded above; their nonzero supports have order type at most ω when listed in decreasing order.

2.3 A standard additive bound

Lemma 2.1. *For surreal x, y ,*

$$\mathbf{b}(-x) = \mathbf{b}(x), \quad \mathbf{b}(x + y) \leq \mathbf{b}(x) \oplus \mathbf{b}(y).$$

Consequently, $\mathbf{b}(\sum_{j=1}^n x_j) \leq \bigoplus_{j=1}^n \mathbf{b}(x_j)$.

Proof. Negation reverses every sign and preserves the length. For addition, use canonical options, whose sign sequences are proper prefixes of the number's sequence. Induct on $b(x) \oplus b(y)$. Every option in the defining cut for $x + y$ has the form $x' + y$ or $x + y'$, where the replaced factor has smaller birthday. By induction and strict monotonicity of natural addition, every such option has birthday strictly below $b(x) \oplus b(y)$. A number defined by a cut has birthday at most $\sup\{b(z) + 1 : z \text{ is an option}\}$. Taking this supremum proves the claimed weak inequality. The finite version follows by induction on n . $\square$

This is the usual additive birthday theorem, not a new result; see [1, p. 96].

3 Real coefficients and sign-expansion tools

3.1 Birthdays of real numbers

Write $d(r) = b(r)$ for $r \in \mathbb{R}$. A real is *dyadic* if it belongs to $\mathbb{Z}[1/2] = \mathbb{Z}[1/2]$; zero is dyadic. Nonzero dyadics have finite birthdays, whereas nondyadic reals have birthday ω . More explicitly, for $r \neq 0$ dyadic with reduced denominator 2^h ,

$$d(r) = \lceil |r| \rceil + h. \quad (3.1)$$

Thus $d(\pm 1) = 1$, $d(\pm \frac{1}{2}) = 2$, $d(\pm \frac{3}{4}) = 3$, and $d(\frac{1}{3}) = \omega$. Formula (3.1) follows from the finite binary sign expansion: for positive noninteger r , it starts with $\lfloor r \rfloor + 1$ plus signs and has h further signs. Negation reverses the signs. The formula also covers integers, for which $h = 0$.

Lemma 3.1 (Real scalar product bound). *For all real r, s ,*

$$d(rs) \leq d(r) \otimes d(s).$$

Proof. Zero factors are immediate. If at least one nonzero factor is nondyadic, the right side is at least ω , whereas rs is real and has birthday at most ω . Suppose both factors are dyadic. Put $A = \lceil |r| \rceil$, $B = \lceil |s| \rceil$, and let $2^h, 2^k$ be their reduced denominators. Then $A, B \geq 1$, the reduced denominator exponent of rs is at most $h + k$, and $\lceil |rs| \rceil \leq AB$. Hence

$$d(rs) \leq AB + h + k \leq (A + h)(B + k) = d(r) d(s).$$

Natural multiplication of finite ordinals is ordinary integer multiplication. $\square$

3.2 The sign-expansion input

We use Gonshor's sign-expansion theorem in the formulation restated by Ehrlich [4, Propositions 3.7–3.8]; see also [5, Definition 2.6 and Theorem 2.7]. Here is the version needed for our calculations.

If a has a sign sequence and p_δ denotes the number of plus signs strictly before its δ th sign, the sequence for ω^a starts with a plus. For each successive sign of a , append a block of that sign of length $\omega^{p_\delta+1}$. For $r > 0$, append the signs of r after its first plus, repeating each of them ω^p times, where p is the total number of pluses in a . Negative r reverses the resulting signs.

For a normal form $\sum_i r_i \omega^{a_i}$, first *reduce* each exponent's sign sequence. A minus is deleted if its entire prefix, including that minus, already occurs in an earlier exponent. There is a second rule: if

the current exponent starts with the immediately preceding exponent followed by a minus, and the preceding coefficient is nondyadic, delete that additional minus. Compute a term block from each reduced exponent by the preceding rule, then concatenate the blocks. No plus signs are deleted.

Lemma 3.2 (Prefixes and countable unions). *A normal-form truncation is a sign-sequence prefix, and therefore has birthday at most that of the complete normal form. If a normal form has infinitely many terms indexed by $\mathbb{N}$, then*

$$\mathbf{b}\left(\sum_{i<\omega} r_i \omega^{a_i}\right) = \sup_{n<\omega} \mathbf{b}\left(\sum_{i<n} r_i \omega^{a_i}\right). \quad (3.2)$$

Proof. The blocks for an initial normal-form segment are unchanged by later terms: both deletion rules look only at earlier exponents and coefficients. Thus its sign sequence is the corresponding initial concatenation. At a limit of such segments the complete sign sequence is their union, whose length is the supremum of their lengths. $\square$

3.3 The negative-integer block calculation

For an exponent consisting of k minuses, with k a nonnegative integer, let $L(k, r)$ denote the birthday of the term block with nonzero real coefficient r . Here the exponent is a *reduced* exponent; it need not equal the original exponent of a term.

Lemma 3.3. *The exact block lengths are*

$$L(k, r) = \begin{cases} \mathbf{d}(r), & k = 0, \\ \omega \cdot k + (\mathbf{d}(r) - 1), & k \geq 1, r \in \mathbb{Z}[1/2], \\ \omega \cdot (k + 1), & k \geq 1, r \notin \mathbb{Z}[1/2]. \end{cases} \quad (3.3)$$

Proof. There are no plus signs in the exponent, so each exponent sign contributes ω signs, and coefficient signs contribute one sign each. If $k \geq 1$, the initial plus is absorbed in ordinary ordinal addition: $1 + \omega \cdot k = \omega \cdot k$. A dyadic coefficient contributes precisely $\mathbf{d}(r) - 1$ further signs. A nondyadic coefficient contributes ω further signs. If $k = 0$, only the real coefficient's sign sequence remains. Negating r changes no length. $\square$

4 An exact case with ordinal exponents

Consider the class ring

$$O = \left\{ \sum_{i=1}^m r_i \omega^{\alpha_i} : m < \omega, \quad \alpha_i \in \mathbf{On}, \quad r_i \in \mathbb{R} \right\}.$$

Zero coefficients are omitted and equal exponents are collected. It is a ring because a product exponent is $\alpha_i \oplus \beta_j$, again an ordinal. Normal forms supported on ordinals are necessarily finite: an infinite support would yield an infinite strictly decreasing sequence of ordinals.

Theorem 4.1 (Ordinal-exponent polynomials). *For $P = \sum_{i=1}^m r_i \omega^{\alpha_i}$ in decreasing normal form,*

$$b(P) = \bigoplus_{i=1}^m (\omega^{\alpha_i} \otimes d(r_i)). \quad (4.1)$$

For all $P, Q \in O$,

$$b(PQ) \leq b(P) \otimes b(Q).$$

Proof. An ordinal exponent has only plus signs, so neither deletion rule applies. Since ω^α is the corresponding ordinal, its birthday is ω^α . The coefficient part of the sign expansion gives

$$b(r\omega^\alpha) = \begin{cases} \omega^\alpha \cdot d(r), & r \in \mathbb{Z}[1/2] \setminus \{0\}, \\ \omega^{\alpha+1}, & r \notin \mathbb{Z}[1/2]. \end{cases}$$

Both cases equal $\omega^\alpha \otimes d(r)$.

The birthday of the whole normal form is the ordinary concatenation sum of these ordinal monomials. Their Cantor exponents are α_i or $\alpha_i + 1$. If $\alpha_i > \alpha_j$ then $\alpha_i \geq \alpha_j + 1$, so those Cantor exponents are nonincreasing in normal-form order. Consequently the ordinary sum equals the natural sum, proving (4.1).

For two individual terms, Lemma 3.1 and (2.1) give

$$\begin{aligned} b((r\omega^\alpha)(s\omega^\beta)) &= \omega^{\alpha \oplus \beta} \otimes d(rs) \\ &\leq (\omega^\alpha \otimes d(r)) \otimes (\omega^\beta \otimes d(s)). \end{aligned}$$

Expand PQ into its finitely many pairwise term products and apply Lemma 2.1. Natural distributivity and (4.1) then give the result. Possible equal exponents or coefficient cancellations can only improve the resulting upper bound; the additive lemma does not require the expanded sum to be in normal form. $\square$

Example 4.2. Let $P = 3\omega^2 + \frac{1}{3}\omega - \frac{1}{2}$. Its term birthdays are $\omega^2 \cdot 3$, ω^2 , and 2, respectively. Hence

$$b(P) = \omega^2 \cdot 4 + 2.$$

The coefficient at exponent 1 contributes at Cantor exponent 2 because it is nondyadic. This shift is why a degree estimate must distinguish dyadic and nondyadic leading coefficients.

5 The exact finite Laurent birthday formula

Let $\mathcal{F} = \mathbb{R}[\omega, \omega^{-1}]$. For $x \in \mathcal{F}$, write

$$x = P + u, \quad P = x_{>0} = \sum_{k \geq 1} c_k \omega^k, \quad u = x_{\leq 0} = \sum_{k \leq 0} c_k \omega^k.$$

Both sums are finite. The term *bounded* means that the element lies between two real numbers; for a Laurent polynomial this is equivalent to having no positive exponent. Put

$$H(P) = \bigoplus_{k \geq 1} (\omega^k \otimes d(c_k)) = b(P), \quad (5.1)$$

where zero coefficients contribute zero.

Theorem 5.1 (Finite Laurent formula). *If x has no negative-exponent term, then*

$$b(x) = H(P) \oplus d(c_0).$$

Otherwise let $-m$, $m \geq 1$, be its lowest nonzero exponent and put $a = c_{-m} \neq 0$. If $a \notin \mathbb{Z}[1/2]$, then

$$b(x) = H(P) + \omega \cdot (m + 1). \quad (5.2)$$

If $a \in \mathbb{Z}[1/2]$, define

$$\varepsilon = \begin{cases} 1, & c_{-m+1} \notin \mathbb{Z}[1/2], \\ 0, & c_{-m+1} \in \mathbb{Z}[1/2], \end{cases}$$

with an absent coefficient interpreted as zero. Then

$$b(x) = H(P) + \omega \cdot m + (d(a) - 1 + \varepsilon). \quad (5.3)$$

All unadorned plus signs on these right sides are ordinary ordinal additions.

Proof. First compute the birthday of the bounded part u alone. Its negative exponents have sign sequences consisting entirely of minuses. Suppose the current term is $c\omega^{-m}$ and the preceding nonzero term of the bounded part has exponent $-p$, where $0 \leq p < m$. Its sign sequence has p minuses, so the first deletion rule removes exactly those p shared minuses. Put $\chi = 1$ if that preceding coefficient is nondyadic and $\chi = 0$ otherwise. The second rule removes one additional minus precisely when $\chi = 1$. Thus the current reduced exponent has

$$k = m - p - \chi \quad (5.4)$$

minuses. This is nonnegative because $m - p \geq 1$.

Before the first negative term, there may be a constant coefficient. In that case the same description applies with $p = 0$. If there is no constant term, use an empty preceding part, $p = 0$, $\chi = 0$, and length zero.

We prove the claimed bounded-part formula inductively along its nonzero negative terms. If the preceding coefficient is nondyadic, the preceding length is $\omega \cdot (p + 1)$. If it is dyadic, the preceding length is $\omega \cdot p + t$ for a finite t ; for an initial dyadic constant this simply means its finite birthday. These statements are also valid for the empty initial part with $p = t = 0$.

Suppose first that $k \geq 1$. Appending the block from Lemma 3.3 erases that finite t and gives an ω coefficient of

$$(p + \chi) + k = m$$

when c is dyadic, or $m + 1$ when c is nondyadic. In the dyadic case the finite remainder is $d(c) - 1$. In the nondyadic case it is zero.

The exceptional case is $k = 0$. Equation (5.4) then forces $m = p + 1$ and $\chi = 1$: the previous coefficient is nondyadic at the immediately adjacent exponent. The preceding length is $\omega \cdot m$, and the current block has length $d(c)$. For dyadic c , the new length is therefore $\omega \cdot m + d(c)$, one larger in its finite remainder than the usual formula. For nondyadic c , it is $\omega \cdot (m + 1)$. This proves the asserted ε correction and completes the induction.

It remains to restore P . Positive integer exponents have only plus signs. They neither share a minus-containing prefix with a nonpositive exponent nor trigger the second deletion rule across the positive/bounded boundary. Consequently the sign sequence of $P + u$ is the sign sequence of P followed by exactly the same blocks as for u alone. Its length is $H(P) + b(u)$. If u is real, Theorem 4.1 gives the stated nonnegative formula. Otherwise the calculation just made gives (5.2) and (5.3). $\square$

Corollary 5.2 (Exact finite splitting). *For finite Laurent polynomials $x = P + u$ as above,*

$$b(x) = b(P) \oplus b(u). \quad (5.5)$$

Every bounded finite Laurent polynomial has birthday below ω^2 . If it has a negative-exponent term, its birthday is at least ω .

Proof. The finite formula gives $b(u) = \omega \cdot M + \tau$ with M, τ finite, or a real birthday at most ω . On the other hand, every nonzero Cantor term of $H(P)$ has exponent at least 1. Hence concatenating $b(u)$ after $H(P)$ discards no nonzero Cantor term of $H(P)$. Ordinary addition and natural addition agree in this specific configuration. The other assertions follow immediately from the formula. $\square$

5.1 Why the boundary correction matters

The exact value of the coefficient immediately above the lowest exponent is irrelevant, except for whether it is nondyadic. For example,

$$\begin{aligned} b(1 + \omega^{-1}) &= \omega, & b(\tfrac{1}{3} + \omega^{-1}) &= \omega + 1, \\ b(\tfrac{1}{3} + \omega^{-2}) &= \omega \cdot 2, & b(\tfrac{1}{3}\omega^{-1} + \tfrac{1}{2}\omega^{-2}) &= \omega \cdot 2 + 2. \end{aligned}$$

In the last example, the second negative exponent is reduced to the empty sequence, so its coefficient contributes two signs rather than one. A rule that always appends $d(a) - 1$ signs would miss this case. A gap of two or more integer exponents prevents this finite correction from surviving, although the nondyadic deletion still takes place within the intermediate calculation.

6 The finite product theorem

The exact splitting formula reduces the problem to three products: positive parts with positive parts, positive parts with bounded parts, and bounded parts with bounded parts. The first was covered by Theorem 4.1.

6.1 Bounded factors

Lemma 6.1. *For bounded finite Laurent polynomials u, v ,*

$$b(uv) \leq b(u) \otimes b(v).$$

If u has a negative-exponent term and $v \notin \{0, 1, -1\}$, this inequality is strict.

Proof. Zero factors cause no difficulty. If both factors are real, apply Lemma 3.1. If both have negative-exponent terms, then

$$b(uv) < \omega^2 \leq b(u) \otimes b(v),$$

since their product is still a bounded finite Laurent polynomial and both factor birthdays are at least ω .

It remains to multiply a bounded finite Laurent polynomial u with a genuine negative term by a nonzero real r . Write

$$b(u) = \omega \cdot M + \tau, \quad M \geq 1, \quad M, \tau < \omega.$$

If r is nondyadic, the proposed bound is at least ω^2 , while $b(ru) < \omega^2$. If r is dyadic, put $d = d(r)$. Multiplication by a nonzero real leaves the lowest exponent of u unchanged. A dyadic lowest coefficient stays dyadic; a nondyadic one can stay nondyadic or become dyadic. By the exact formula, the leading ω coefficient of $b(ru)$ is thus at most M . For $d \geq 2$, the proposed bound has leading coefficient $dM > M$:

$$b(u) \otimes d = \omega \cdot (dM) + d\tau.$$

This proves strictness. The only nonzero reals with $d = 1$ are $r = \pm 1$; then $b(ru) = b(u)$. $\square$

6.2 A coarse upper bound with one sharp edge case

Lemma 6.2. *Let z be a finite Laurent polynomial with all exponents at most n , where $n \geq 1$ is an integer. Then*

$$b(z) < \omega^{n+1} \cdot 2.$$

More precisely, if the coefficient at exponent n is dyadic, including zero, then $b(z) < \omega^{n+1}$. If it is nondyadic, then

$$b(z) = \omega^{n+1} + \rho \quad \text{for some } \rho < \omega^{n+1}.$$

Proof. By Theorem 5.1, the positive part contributes at most one term at Cantor exponent $n + 1$, arising only from a nondyadic coefficient at exponent n . That term has coefficient one. All remaining positive-power contributions have Cantor exponent at most n , and the bounded contribution is below ω^2 . Since $n \geq 1$, their finite natural sum is below ω^{n+1} . $\square$

Lemma 6.3 (Positive monomial times bounded part). *Let $n \geq 1$ be an integer, $r \neq 0$ real, and u a bounded finite Laurent polynomial. Then*

$$b(r\omega^n u) \leq b(r\omega^n) \otimes b(u). \quad (6.1)$$

If u has a negative-exponent term, the inequality is strict.

Proof. If u is real, both factors belong to $\mathcal{O}$, and Theorem 4.1 applies. Otherwise write $b(u) = \omega \cdot M + \tau$, with $M \geq 1$ and M, τ finite. The product $r\omega^n u$ has highest exponent at most n .

If r is nondyadic, then $b(r\omega^n) = \omega^{n+1}$, and the right side of (6.1) has leading Cantor exponent $n + 2$. Lemma 6.2 bounds the left side by an ordinal with leading exponent at most $n + 1$, so strictness follows.

Suppose that r is dyadic and write $d = d(r)$. The proposed bound is

$$\omega^{n+1} \cdot (dM) + \omega^n \cdot (d\tau). \quad (6.2)$$

If $dM \geq 2$, Lemma 6.2 makes the left side strictly smaller already. Only the case $d = M = 1$ remains.

Here $r = \pm 1$. The finite formula forces the lowest exponent of u to be -1 , with a nonzero dyadic coefficient a . Thus

$$u = c + a\omega^{-1}, \quad c \in \mathbb{R}, \quad a \in \mathbb{Z}[1/2] \setminus \{0\}.$$

If c is dyadic, the top coefficient of $r\omega^n u$ is dyadic, so the left side is below ω^{n+1} and the result follows. If c is nondyadic, the boundary correction is indispensable:

$$b(u) = \omega + d(a).$$

The ordinal-exponent formula gives

$$\begin{aligned} b(r\omega^n u) &= \omega^{n+1} + \omega^{n-1} \cdot d(a), \\ b(r\omega^n) \otimes b(u) &= \omega^{n+1} + \omega^n \cdot d(a). \end{aligned}$$

Since $n \geq 1$ and $d(a) > 0$, the first is strictly smaller. This includes $n = 1$, when the lower term on the first line is finite. $\square$

Corollary 6.4. *For a positive-power polynomial P and a bounded finite Laurent polynomial u ,*

$$b(Pu) \leq b(P) \otimes b(u).$$

If $P \neq 0$ and u has a negative-exponent term, the inequality is strict.

Proof. Expand P into its finitely many nonzero terms. Apply Lemma 2.1 to the products, Lemma 6.3 to each term, and Theorem 4.1 to recombine the birthdays of P by natural addition. For strictness, at least one strict term inequality occurs, and natural addition is strictly increasing. $\square$

6.3 Putting the four products together

Theorem 6.5 (Finite Laurent product theorem). *For all $x, y \in \mathcal{F}$,*

$$b(xy) \leq b(x) \otimes b(y).$$

Proof. Write $x = P + u$ and $y = Q + v$ with positive-power polynomials P, Q and bounded parts u, v . Expand

$$xy = PQ + Pv + Qu + uv.$$

By the additive birthday lemma and the three product estimates,

$$\begin{aligned} b(xy) &\leq b(PQ) \oplus b(Pv) \oplus b(Qu) \oplus b(uv) \\ &\leq (b(P) \otimes b(Q)) \oplus (b(P) \otimes b(v)) \\ &\quad \oplus (b(u) \otimes b(Q)) \oplus (b(u) \otimes b(v)) \\ &= (b(P) \oplus b(u)) \otimes (b(Q) \oplus b(v)) \\ &= b(x) \otimes b(y), \end{aligned}$$

where the last equality is the exact splitting formula (5.5). This proof does not assume that any of the four product summands has a disjoint support from the others. $\square$

Corollary 6.6 (Equality with a finite negative tail). *If $x \in \mathcal{F}$ has a negative-exponent term, then for $y \in \mathcal{F}$,*

$$b(xy) = b(x) \otimes b(y) \iff y \in \{0, 1, -1\}.$$

Proof. The three displayed values give equality directly. Otherwise, in the preceding proof u has a negative-exponent term. If $Q \neq 0$, the Qu estimate is strict by Corollary 6.4. If $Q = 0$, then $y = v$ is bounded and is not 0 or ± 1 , so the uv estimate is strict by Lemma 6.1. Strict monotonicity of the finite natural sum carries that strictness to the complete bound. $\square$

7 Infinite Laurent series and finite jets

7.1 The field and its product law

With $t = \omega^{-1}$, the field in (1.3) is the usual formal Laurent series field $\mathbb{R}((t))$ embedded by normal forms. For completeness, its field property can be seen directly. Every nonzero element has the shape

$$c t^m (1 + h), \quad c \in \mathbb{R} \setminus \{0\}, \quad m \in \mathbb{Z}, \quad h \in t\mathbb{R}[[t]].$$

Its inverse is the formal series

$$c^{-1} t^{-m} \sum_{j \geq 0} (-h)^j.$$

For any fixed power of t , only finitely many terms contribute, since every term of h has exponent at least one. The coefficientwise geometric-series identity proves that this is an inverse.

In omega-exponent notation, if x and y have highest exponents N_x and N_y , the coefficient at exponent k in their product is

$$[\omega^k](xy) = \sum_{i+j=k} c_i d_j.$$

This sum is finite: the possible i satisfy $k - N_y \leq i \leq N_x$. The argument uses formal coefficients only, not analytic convergence.

7.2 Exact birthdays of infinite tails

Theorem 7.1 (Infinite Laurent birthday formula). *Suppose $x \in \mathcal{L}$ has infinitely many nonzero terms. Let $P = x_{>0}$ and $H = b(P)$. Then*

$$\boxed{b(x) = H + \omega^2}, \tag{7.1}$$

with ordinary ordinal addition. In particular, every bounded infinite Laurent series has birthday exactly ω^2 .

Proof. The positive part is finite. List the nonzero negative exponents as $-m_1 > -m_2 > \dots$. The positive integers m_j tend to infinity, since the support is infinite and integer-valued. Let x_j be the finite normal-form truncation through exponent $-m_j$. Theorem 5.1 gives

$$b(x_j) = H + \omega \cdot M_j + \tau_j,$$

where M_j equals m_j or $m_j + 1$, and τ_j is finite. Consequently these lengths are cofinal in $H + \omega^2$. The sign sequences are nested prefixes by Lemma 3.2, and their union is the sign sequence of x . Taking their supremum proves (7.1). $\square$

Remark 7.2 (A real loss of a Cantor term). Let $x = \omega + \sum_{n \geq 1} \omega^{-n}$. Then

$$b(x) = \omega + \omega^2 = \omega^2, \quad b(\omega) \oplus b\left(\sum_{n \geq 1} \omega^{-n}\right) = \omega^2 + \omega.$$

Thus the finite splitting equality (5.5) does not extend unchanged to infinite tails. The difference is not a notation preference: it is the ordinary-ordinal absorption of a genuine ω term.

7.3 Stabilization of a finite product truncation

For an integer K , define

$$x_{\geq K} = \sum_{k \geq K} c_k \omega^k.$$

This is a finite normal-form truncation for every $x \in \mathcal{L}$.

Lemma 7.3 (Finite-jet stabilization). *Let $x, y \in \mathcal{L} \setminus \{0\}$ have highest exponents N_x, N_y . Fix $K \in \mathbb{Z}$ and choose any integer T such that*

$$T \leq K - N_y, \quad T \leq K - N_x. \quad (7.2)$$

Then

$$(x_{\geq T} y_{\geq T})_{\geq K} = (xy)_{\geq K}. \quad (7.3)$$

Proof. A discarded term of x has exponent $i < T$. Multiplied by any term of y , it has exponent $i + j < T + N_y \leq K$. Likewise, a discarded term of y cannot contribute to exponent K or above. Therefore the complete products have the same coefficients at every exponent at least K . These coefficients are finite sums, so there is no exchange of infinite summation operations hidden in the argument. $\square$

Proof of Theorem 1.2. Zero factors are immediate. Fix nonzero $x, y \in \mathcal{L}$ and put

$$B = b(x) \otimes b(y).$$

For any integer cutoff K , choose T as in Lemma 7.3. Since normal truncations are sign prefixes, and since $x_{\geq T}, y_{\geq T}$ are finite Laurent polynomials, we have

$$\begin{aligned} b((xy)_{\geq K}) &= b((x_{\geq T} y_{\geq T})_{\geq K}) \\ &\leq b(x_{\geq T} y_{\geq T}) \\ &\leq b(x_{\geq T}) \otimes b(y_{\geq T}) \\ &\leq b(x) \otimes b(y) = B. \end{aligned}$$

The middle inequality is Theorem 6.5, and the last one is monotonicity of natural multiplication.

If xy has finite support, choose K below its lowest exponent, and this is the desired bound. If it has infinite support, the finite truncations exhaust its sign sequence by Lemma 3.2. Their birthdays are all bounded by the *same* ordinal B , so their supremum is bounded by B as well. Thus $b(xy) \leq B$ in either case. $\square$

The proof deliberately does not assert a general continuity identity such as $\sup_n (\alpha_n \otimes \beta_n) = (\sup_n \alpha_n) \otimes (\sup_n \beta_n)$. It only keeps one fixed upper bound throughout. Even natural addition need not preserve a supremum in a variable: $\sup_{n < \omega} (n \oplus 1) = \omega$, whereas $(\sup_{n < \omega} n) \oplus 1 = \omega + 1$.

8 Examples and exact consequences

8.1 A table of birthdays

The following values follow directly from the finite and infinite formulas. The sums displayed here are surreal sums; every expression in the right column is an ordinal in Cantor normal form.

Surreal number	Birthday
$1 + \omega^{-1}$	ω
$\frac{1}{3} + \omega^{-1}$	$\omega + 1$
$\omega + \omega^{-1}$	$\omega \cdot 2$
$\omega^{-m}, m \geq 1$	$\omega \cdot m$
$\frac{1}{2}\omega^{-m}, m \geq 1$	$\omega \cdot m + 1$
$\frac{1}{3}\omega^{-m}, m \geq 1$	$\omega \cdot (m + 1)$
$\frac{1}{3}\omega^{-1} + \frac{1}{2}\omega^{-2}$	$\omega \cdot 2 + 2$
$\sum_{n \geq 1} \omega^{-n}$	ω^2
$\omega + \sum_{n \geq 1} \omega^{-n}$	ω^2
$\omega^3 + \sum_{n \geq 1} \omega^{-n}$	$\omega^3 + \omega^2$
$\frac{1}{3}\omega^2 + \sum_{n \geq 1} \omega^{-n}$	$\omega^3 + \omega^2$
$\sum_{n \geq 0} \omega^{-n}/n!$	ω^2

8.2 How loose can the natural-product bound be?

For positive integers m, n ,

$$\mathbf{b}(\omega^{-m}\omega^{-n}) = \omega \cdot (m + n), \quad \mathbf{b}(\omega^{-m}) \otimes \mathbf{b}(\omega^{-n}) = \omega^2 \cdot mn.$$

The conjectured bound is therefore already very non-sharp for elementary infinitesimals. Cancellation can make the discrepancy larger still. If

$$x = \sum_{n \geq 0} \omega^{-n}, \quad y = 1 - \omega^{-1},$$

then $xy = 1$, but

$$\mathbf{b}(x) = \omega^2, \quad \mathbf{b}(y) = \omega, \quad \mathbf{b}(x) \otimes \mathbf{b}(y) = \omega^3.$$

The theorem concerns a uniform complexity bound, not an estimate expected to be nearly an equality for typical non-Archimedean factors.

8.3 Reciprocals of finite Laurent polynomials

Corollary 8.1. *Let $f \in \mathcal{F} \setminus \{0\}$ have at least two nonzero terms and highest exponent $N \geq 0$. Then*

$$b(f^{-1}) = \omega^2. \quad (8.1)$$

Proof. The only units in the finite Laurent polynomial ring over a field are the nonzero monomials. Indeed, for a nonzero finite Laurent polynomial define its width as highest exponent minus lowest exponent. The extreme coefficients of a product are nonzero products of extreme coefficients, so widths add under multiplication. A product equal to 1 has width zero, forcing both factors to have width zero.

Thus f^{-1} is not finite; it has infinitely many terms in $\mathcal{L}$. Its highest exponent is $-N \leq 0$, so it is bounded. Apply Theorem 7.1. $\square$

For example,

$$(\omega + 1)^{-1} = \omega^{-1} - \omega^{-2} + \omega^{-3} - \dots = \sum_{n \geq 1} (-1)^{n-1} \omega^{-n}$$

where the displayed identity is coefficientwise formal multiplication; its birthday is ω^2 . Likewise, $(1 + \omega^{-1})^{-1}$ and $(\omega^2 + \omega)^{-1}$ have birthday ω^2 . The assumption $N \geq 0$ cannot be silently omitted: a reciprocal with positive leading exponent can have a positive polynomial part that survives in $H + \omega^2$.

8.4 The complete birthday spectrum of this field

Corollary 8.2. *The set of birthdays realized in $\mathcal{L}$ is exactly*

$$\{\alpha : \alpha < \omega^\omega\}.$$

Proof. The finite and infinite formulas place every birthday below ω^ω . Conversely, every ordinal $\alpha < \omega^\omega$ has a finite Cantor normal form with finite exponents and nonnegative integer coefficients. It is itself an element of $\mathcal{L}$, and $b(\alpha) = \alpha$. $\square$

This does not identify $\mathcal{L}$ with the entire set of surreals of birthday below ω^ω . For instance, $\omega^{1/2}$ has birthday ω^2 but is not in $\mathcal{L}$, because its unique normal form has a noninteger exponent. In particular $\mathcal{L}$ is not real closed: its positive element ω has no square root in $\mathcal{L}$.

9 What the proof does not extend to automatically

9.1 The failed termwise shortcut

The known monomial case might suggest expanding every product into term products and summing their birthday bounds. This gives

$$b(xy) \leq \left(\bigoplus_i b(r_i \omega^{a_i}) \right) \otimes \left(\bigoplus_j b(s_j \omega^{b_j}) \right)$$

for finite normal forms, but does not in general give the desired bound in terms of $b(x)$ and $b(y)$. One would need equality between a number's birthday and the natural sum of its term birthdays, which can fail.

The simplest Laurent example is

$$b(1 + \omega^{-1}) = \omega < \omega + 1 = b(1) \oplus b(\omega^{-1}).$$

Even with positive fractional exponents,

$$b(\omega + \omega^{1/2}) = \omega^2, \quad b(\omega) \oplus b(\omega^{1/2}) = \omega^2 + \omega.$$

Here $1/2$ has sign sequence $+-$, so $\omega^{1/2}$ has a first block of ω pluses followed by ω^2 minuses. Its birthday is ω^2 . The leading ω term of the sum is absorbed in the concatenated birthday.

The positive/bounded decomposition in the finite Laurent proof was chosen precisely to retain exact equality at the recombination step. It is not a claim that every normal-form decomposition has that property.

9.2 Fractional dyadic exponents

A natural next restriction is

$$\mathbb{R}[\omega^q : q \in \mathbb{Z}[1/2]],$$

where each expression has finite support. The general sign-expansion algorithm still works with finite exponent sign sequences, so exact computations are feasible. What is missing is an analogue of the finite Laurent formula that retains the sharp birthday budget under arbitrary fractional-exponent products.

The obstacle is structural. Positive integer exponents have no minuses, while negative integer exponents lie on a single all-minus branch. This made the reduction rule depend only on the previous negative exponent and one coefficient flag. A general dyadic exponent can have an arbitrary finite mixture of signs. An earlier, nonadjacent exponent can then delete a minus in a later exponent, and multiple sign-block scales can interact. The last exponent and its immediate predecessor no longer summarize all relevant information.

The accompanying search found no counterexample in 100,000 random pairs from a specified dyadic grid, but this is not a proof of that larger restriction. A useful next technical target is a *prefix-sensitive* bound on the reduced term blocks: it must account for the finite tree of shared exponent prefixes, rather than merely adding independent term lengths.

9.3 Infinite supports beyond the integer lattice

The finite-jet argument used that every cutoff $x_{\geq K}$ is finite. This is not true for arbitrary reverse-well-ordered supports in a dense exponent group. For example, the descending sequence

$$-\frac{1}{2}, -\frac{3}{4}, -\frac{7}{8}, -\frac{15}{16}, \dots$$

is a legitimate dyadic support, but all its infinitely many terms lie above -1 . Thus the same numerical cutoff no longer produces a finite polynomial. This example does not disprove a more general approximation theorem; it marks the specific step that has not been justified outside the discrete Laurent setting.

9.4 No general solution has been obtained

The full conjecture permits arbitrary surreal exponents, arbitrary set-sized reverse-well-ordered supports, and products for which both sign reductions and support order types are much more complicated. Neither the polynomial theorem with ordinal exponents nor the Laurent-field theorem covers those cases. The computational search supplies evidence, not the missing argument.

10 Exact computation and reproducibility

The archive includes `verify.py`, a standard-library-only Python program compatible with Python 3.9 or later, and the recorded results in `verification_results.json`. Mathematical arithmetic is exact: coefficients and dyadic exponents use `fractions.Fraction`; ordinal operations use integer coefficient tuples. No floating-point approximation enters a birthday comparison.

10.1 Two independent routes to finite Laurent birthdays

The function `sign_birthday` implements the general finite-exponent sign-expansion theorem. For each exponent it records its finite sign sequence, removes already-shared minus prefixes, applies the separate nondyadic predecessor deletion, and concatenates ordinal block lengths.

The function `laurent_formula` instead implements Theorem 5.1, using only positive-power term birthdays, the lowest exponent and coefficient, and the single boundary flag. It does not call the sign-expansion engine. Agreement between the two implementations is a check on the algebraic simplification and on edge-case handling, not an independent proof of Gonshor’s sign-expansion theorem.

For instance, the tuple $(2, 4, 0, 1)$ represents $\omega^3 + \omega \cdot 4 + 2$. Natural addition is componentwise addition, natural multiplication is convolution, and ordinary addition discards every coefficient of the left operand below the leading exponent of the right operand before combining the leading coefficients. Implementing both additions separately is essential to the tests.

10.2 Recorded verification

The completed checks are summarized below. Their precise grids, seed, and sampling list are recorded in the JSON file.

Check	Cases	Result
Finite Laurent formula versus sign expansion	117,649	All equal
Finite Laurent product inequality, ordered pairs	15,625	All passed
Strictness with a negative tail, within those pairs	12,200	All strict
Random dyadic-exponent product inequality	100,000	All passed
Dyadic real length versus actual sign length	405	All equal
Named birthday regressions	12	All passed
Ordinal arithmetic regressions	3	All passed

The formula test uses exponents $-3, -2, -1, 0, 1, 2$ and coefficient choices

$$\{0, 1, -1, \frac{1}{2}, 2, \frac{1}{3}, -\frac{1}{3}\},$$

so it tests all $7^6 = 117,649$ patterns, including zero coefficients and missing-predecessor cases. The product test uses all $5^3 = 125$ polynomials on exponents $-1, 0, 1$ with coefficients in $\{0, 1, -1, \frac{1}{2}, \frac{1}{3}\}$, and tests all ordered pairs. Exactly 881 pairs attain equality; the strictness test concerns the specified 12,200-pair subset.

The exploratory search uses exponents $i/8$ for $-24 \leq i \leq 24$, one through seven distinct exponents per factor, and rational coefficients drawn from the ordered sampling list generated by

$$i/j, \quad j \in (1, 2, 4, 3, 5), \quad i \in (-3, -2, -1, 1, 2, 3).$$

Repeated rational values in this list are intentionally retained. The random seed is 394230. Among its 100,000 pairs, 202 attain equality. This search is broader than the theorem in its allowed finite exponents, but much narrower in its coefficients and support sizes.

10.3 Running the checks

From the extracted archive directory, run

```
python verify.py --output verification_results.json
```

A smaller smoke test is available as

```
python verify.py --quick --output quick_results.json
```

The output records the Python version and run date. Wall-clock duration is machine-dependent. Reproduction should compare the mathematical counts and outcomes, not timestamps or timing fields.

The program is not a formal verification of the proofs. It cannot quantify over arbitrary real coefficients, infinite series, or all ordinals. The argument for those cases is the mathematical one in the article, with its external inputs explicitly identified.

11 Proof audit and conclusion

The proof has two nontrivial interfaces to standard surreal theory: the normal-form/Hahn-series correspondence, and the sign-expansion theorem with its two deletion rules. The additive birthday bound was recalled with its usual recursive proof. No instance of the conjectured general multiplicative bound is assumed in proving the Laurent case.

The internal chain of implications is

$$\begin{aligned} \text{sign expansions} &\implies \text{exact finite Laurent birthdays} \\ &\implies \text{bounded and cross-product estimates} \\ &\implies \text{finite Laurent product bound} \\ &\implies \text{fixed-bound finite-jet argument} \\ &\implies \text{full Laurent-series product bound.} \end{aligned}$$

Several potential gaps were checked explicitly. Negative coefficients reverse signs but not lengths. An absent coefficient is dyadic zero, so it must not activate the boundary correction. Adjacent nondyadic coefficients require the second deletion rule even though the shared-prefix rule has already been applied. Multiplying by a dyadic scalar can turn a nondyadic coefficient into a dyadic one, but cannot turn a dyadic coefficient into a nondyadic one. The only sharp cross-product case has a coefficient of birthday one and a bounded tail with leading birthday coefficient one; it was treated separately. Finally, infinite support was handled by a uniform bound on finite product truncations, not by an unproved continuity property of natural operations.

The resulting affirmative restriction is the entire field $\mathbb{R}((\omega^{-1}))$, not merely a collection of monomials or rational functions. The exact finite and infinite birthday formulas supply additional information that is independent of the product inequality. In particular they expose why some straightforward-looking normal-form arguments lose the required bound, and they identify a concrete next obstacle: the shared-prefix combinatorics of fractional dyadic exponents. This is the extent of the result established in this article; the full birthday product conjecture remains beyond this proof.

A Bibliographic and provenance notes

The research status discussion is deliberately narrower than a claim that a complete literature survey has proved absence of later work. The 2001 source contains the monomial theorem and the general weaker estimate; the 2024 statement is first-person testimony by one of its authors. The 2011 paper and 2022 preprint provide accessible formulations of the sign-expansion rules used here. No purported solution of a different conjecture bearing Gonshor’s name was treated as a resolution of the birthday question.

The original paper by van den Dries and Ehrlich has an erratum [3]. The present proof does not invoke their estimates for additive monoids of supports, closure under inversion, or exponential-field constructions. The affirmative monomial theorem is cited for context; our restricted product proof instead uses the elementary real-coefficient lemma and the stated sign rules. This separation keeps the logical dependencies of the argument visible.

The archive contains the article source and PDF, exact verification code, recorded finite checks, a README with build instructions, and an audit file. It does not include third-party books or papers, font files, or a claim of proof-assistant certification.

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